# Priyan Vaithilingam

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## Experience

**Research Scientist, Apple** — *Seattle, USA*
*Apr 2025 - Present*
- Generative UI – on the fly synthesis of UI, adapting to user intent and context.
- Building computer use agents that perceive, reason about, and act on graphical user interfaces.

**Research Intern, Microsoft Research** — *Redmond, USA*
*May 2024 - Aug 2024*
- Embedding AI assistance directly into OS-level interactions like Copy-Paste.
- Created MagicCopy – to helps users smartly copy-paste data between applications using LLMs.

**Research Intern, Microsoft Research** — *Redmond, USA*
*May 2023 - Aug 2023*
- Re-imagining interaction techniques for AI-assisted data visualization tools
- Focusing on dynamic UI for editing and interacting with visualizations that adapts to user intent.

**Research Intern, Microsoft** — *Remote*
*Sept 2021 - Aug 2022*
- Re-imagining interaction techniques for AI-based automated programming tools
- Research on improving Intellicode user experience in Visual Studio

**Research Fellow, Microsoft Research India** — *Bangalore, India*
*Mar. 2016 - July 2018*
- Honorable mention paper in CHI 2018 | Worldwide 2nd place in Microsoft Hackathon 2017.
- Research in Human Computer Interaction, Programming Languages and Information Retrieval.

## Education

Harvard University — *Cambridge, USA*
PhD in Computer Science | CGPA: 4/4 | Advised by Dr. Elena L. Glassman — *July 2020 - March 2025*

University of California, San Diego — *San Diego, USA*
M.S. in Computer Science | GPA: 3.96/4 | Advised by Dr. Philip Guo — *Sept 2018 - June 2020*

Indian Institute of Technology (IIT), Indore — *Indore, India*
B.Tech. in Computer Science & Engineering | GPA: 9/10 | Advised by Dr. Gourinath Banda — *June 2011 - May 2015*

## Publications

**The Way We Notice, That's What Really Matters: Instantiating UI Components with Distinguishing Variations**
CHI 2026
**Priyan Vaithilingam**, Alan Leung, Jeffrey Nichols, Titus Barik

**Semantic Commit: Helping Users Update Intent Specifications for AI Memory at Scale**
UIST 2025
**Priyan Vaithilingam**, Munyeong Kim, Frida-Cecilia Acosta-Parenteau, Daniel Lee, Amine Mhedhbi, Elena L. Glassman, Ian Arawjo

**Imagining a Future of Designing with AI: Dynamic Grounding, Constructive Negotiation, and Sustainable Motivation**
DIS 2024
**Priyan Vaithilingam**, Ian Arawjo, Elena L. Glassman

**DynaVis: Dynamically Synthesized UI Widgets for Visualization Editing**
CHI 2024 (**Best Paper Award**)
**Priyan Vaithilingam**, Elena L. Glassman, Jeevana Priya Inala, Chenglong Wang

**ChainForge: A Visual Toolkit for Prompt Engineering and LLM Hypothesis Testing**
CHI 2024 (**Honorable Mention Award**)
Ian Arawjo, Chelse Swoopes, **Priyan Vaithilingam**, Martin Wattenberg, Elena L. Glassman

**Towards More Effective AI-Assisted Programming: A Systematic Design Exploration to Improve Visual Studio IntelliCode's User Experience**
ICSE SEIP 2023
**Priyan Vaithilingam**, et al.

**Expectation vs. Experience: Evaluating the Usability of Code Generation Tools Powered by Large Language Models**
CHI LBW 2022
**Priyan Vaithilingam**, Tianyi Zhang, Elena L. Glassman

**Assuage: Assembly Synthesis Using A Guided Exploration**
UIST 2021
Jingmei Hu, **Priyan Vaithilingam**, Stephen Chong, Margo Seltzer, Elena L. Glassman

**Interpretable Program Synthesis**
CHI 2021
Tianyi Zhang, Zhiyang Chen, Yuanli Zhu, **Priyan Vaithilingam**, Xinyu Wang, Elena L. Glassman

**Bespoke: Interactively Synthesizing Custom GUIs from Command-Line Applications By Demonstration**
UIST 2019
**Priyan Vaithilingam**, Philip Guo

**CodeTalk: Improving programming environment accessibility for Visually Impaired developers**
CHI 2018 (**Honorable Mention Award**)
Venkatesh Potluri, **Priyan Vaithilingam**, Suresh Iyengar, Vidya Y, Manohar Swaminathan, Gopal Srinivasa

## Patents

**Content Management Tool for Content Transformation**
US Patent 12,705,251 (2026)
J.P. Inala, G.A. Ramos, N.M. Riche, C. Wang, **P. Vaithilingam**

**Software Development Artifact Name Generation**
US Patent 12,487,817 (2025)
P. Groenewegen, **P. Vaithilingam**, R.J. Malpani, W. Wang

**Dynamically Synthesized User Interface Widgets**
US Patent App. 18/533,674 (2025)
J.P. Inala, C. Wang, **P. Vaithilingam**

## Awards

Best Paper Award, CHI 2024 – DynaVis *2024*
Honorable Mention Award, CHI 2024 – ChainForge *2024*
Honorable Mention Award, CHI 2018 – CodeTalk *2018*